Computer Engineering Capstone Project Proposal I
Project AiQ
Intelligence That Drives Execution
An AI-driven Knowledge Management Platform
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2 Abstract

This project, Project AiQ - ”Intelligence That Drives Execution”, addresses the critical problem of fragmented enterprise knowledge scattered across systems like SharePoint, by designing a secure and scalable AI-driven Knowledge Hub to automate retrieval and improve organizational efficiency.

To achieve this, the proposed solution involves designing and implementing a Retrieval-Augmented Generation (RAG) system, specifically focusing on a scalable data ingestion and preparation pipeline that utilizes GraphRAG techniques to securely structure content from enterprise sources.

The expected outcomes include a detailed technical design, an implemented data pipeline with foundational Single Sign-On (SSO) security, a minimal RAG Web Chat MVP,

and significant contributions to both academic validation of the RAG architecture and industrial benefits through improved efficiency and knowledge consolidation.

Keywords: RAG, Knowledge Management, Data Ingestion, Enterprise AI, GraphRAG

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5 1. Introduction & Problem Definition

5.1 1.1 Background & Motivation

Modern enterprise environments, including organizations that manage large and complex projects like **SCB TechX**, often face a common problem: important project information is **scattered across many different systems**. Emails, shared file drives, and internal tools such as SharePoint all contain pieces of knowledge, but none of them provide a complete picture. As a result, finding the right information becomes slow, inconvenient, and sometimes confusing.

To solve this, organizations are moving toward a **Knowledge Hub** concept, which requires a system that not only consolidates data but also provides intelligent access. The central challenge, which motivates this research-focused project, is the initial step: how to securely and efficiently ingest this fragmented data and transform it into a unified, queryable knowledge base. By creating a robust ingestion and preparation pipeline early on, we aim to lay the groundwork for using AI to automate knowledge retrieval, thereby addressing inefficiency issues and ensuring project continuity.

5.2 1.2 Problem Statement

The core engineering problem is to design, research, and validate the technical foundation for a secure and scalable AI-driven knowledge management platform. Specifically, the problem centers on:

1. Designing a robust content extraction and preparation pipeline that can handle data from enterprise-level sources (e.g., SharePoint) and structure it for semantic search.

2. Formulating a security strategy that integrates with existing enterprise authentication methods, such as SSO (Single Sign-On).

3. Integrating the AI and Distributed Systems disciplines to ensure the initial data ingestion architecture is scalable and reliable, which is a complex requirement for real-world enterprise deployment.

This research and design effort is complex and meaningful because successfully managing the ”first mile” of data ingestion and preparation dictates the accuracy and performance of the subsequent RAG pipeline, a non-trivial challenge that requires creative integration of various computer engineering sub-disciplines.

5.3 1.3 Objectives

The project aims to achieve the following clear, measurable goals in a phased, two-semester approach:

1. Research and Design (Capstone I Focus): Conduct comprehensive research on current RAG architectures, perform a gap analysis of existing knowledge management systems, and deliver a detailed technical design for the entire system, emphasizing the data ingestion and content extraction pipeline.

2. Data Preparation Pipeline Development (Implementation Focus): Design and implement a scalable data ingestion and preparation pipeline capable of securely connecting to and extracting content from a primary enterprise source (e.g., SharePoint).

3. Foundational Integration (Implementation Focus): Implement the foundational security integration to support Single Sign-On (SSO), ensuring secure user access to the system.

4. Evaluation: Define key performance metrics for the data pipeline (e.g., latency, throughput) and the RAG model's accuracy, with testing to commence in Capstone II.

5.4 1.4 Scope & Limitations

Scope

1. Research & Proposal (Capstone I): Deliver a full written proposal and research review.
2. Data Ingestion & Preparation: Design and implement the ingestion of unstructured and semi-structured data from SharePoint and a general file upload feature. This includes content extraction, chunking, and embedding.
3. Security Foundation: Implement Single Sign-On (SSO) for secure user authentication.
4. Interface: Develop a minimal Web Chat MVP with basic prompt and response functionality to demonstrate RAG output.

Limitations

1. Initial data ingestion implementation will focus on one primary enterprise source (Xhive).
2. Advanced access control mechanisms (e.g., RBAC) are considered complex and will be scaled back to basic authentication checks (SSO) as advanced access control is considered a future enhancement.
3. Advanced user-facing features (e.g., multi-channel integration via LINE/MS Teams) are out-of-scope for the MVP and reserved for future work.

5.5 1.5 Expected Benefits

The project is structured to maximize academic and professional impact

Academic/Technical Benefit:

1. Validated Architecture: Deliver a thoroughly researched and documented architecture for a complex enterprise RAG system.
2. Data Pipeline Expertise: Provide practical experience in designing and implementing a robust data ingestion pipeline that handles real-world, messy, and large-scale organizational data.
3. RAG Optimization: Provide a testbed for evaluating the performance implications of different RAG components (e.g., chunking strategies, re-ranking).

Industrial/Professional Benefit:

1. Efficiency: Lay the foundation to reduce the time spent searching for information, leading to eventual cost savings and improved team agility.
2. Knowledge Consolidation: Create a centralized platform that turns scattered documentation into an accessible Knowledge Hub.
3. Streamlined Onboarding: Improve the onboarding process for new joiners by providing an immediate, centralized, and intelligent way to access project history and essential internal knowledge.

6 2. Problem Solving & Related Work

6.1 2.1 Review of Existing Knowledge

In large-scale enterprise environments, knowledge is generally locked within different repositories, such as emails, cloud drives, or collaboration platforms, which makes the extraction of information less efficient. The traditional approach to Knowledge Management is based on keyword indexing, such as Inverted Indices, which fail to capture the semantic intent of user queries or contextual relationships between documents [1].

To overcome these limitations, Retrieval-Augmented Generation (RAG) has emerged as a standard architecture. RAG enhances Large Language Models (LLMs) by retrieving relevant data from a private knowledge base (the Vector Store) before generating an answer [2]. The RAG mechanism involves key steps: query transformation, retrieval (using a Vector Store based on semantic similarity), and the final generation step. However, the efficacy of a RAG system is strictly dependent on robust data management strategies, such as text chunking and document ranking [3].

Current academic and industrial research indicates that standard "out-of-the-box" ingestion pipelines often perform naive text chunking (e.g., fixed window size). They slice documents arbitrarily, severing the logical flow and losing critical metadata (such as file hierarchy or authorship). To build a robust Knowledge Hub, modern approaches must utilize High-Fidelity Ingestion Pipelines. These pipelines combine Vector Embeddings (to capture semantic meaning) with Graph/Relational Structures (a technique known as GraphRAG, to map explicit relationships between entities and documents), ensuring that the data fed into the AI is clean, structured, and context-aware. This project emphasizes researching and implementing advanced techniques, including recursive text splitting and Graph-based chunking, to maximize context preservation. [4].

6.2 2.2 Comparison with Existing Systems

The existing solutions of enterprise knowledge retrieval could be grouped into three levels of maturity.

Level 1: Legacy Keyword Search (e.g., Standard Repository Search)

- **Mechanism**: Matches exact keywords in file names or contents.
- **Weakness**: Returns lists of files rather than answers. It cannot handle synonyms or natural language questions
- **Relevance**: Serves as the baseline that this project aims to surpass.

Level 2: Naive/Generic RAG Implementations

- **Mechanism**: Ingests text directly into a vector store using fixed-window chunking (e.g., every 500 tokens).
- **Weakness**: Suffers from Context Fragmentation. Since text chunking often disregards the logical structure, the relationship between topics and content is lost. This prevents the model from correctly linking information and often leads to ”hallucinations” (generating false information) due to the lack of structural integrity in the input data [5].
- **Relevance**: Reflects a critical limitation that this project aims to address by prioritizing the engineering foundation from the very first step (The First Mile of Engineering).

Level 3: The Proposed Pipeline-Centric System (GraphRAG-enhanced)

- **Mechanism**: Prioritizes the Data Ingestion Layer. It employs a custom extraction pipeline (utilizing tools like the Docling framework [6] or similar advanced parsers) that rigorously sanitizes data, preserves document hierarchy, and constructs a hybrid knowledge representation. This representation combines semantic vectors (in Qdrant) with explicit relational context (in a separate Graph Store like Neo4j, or through metadata filtering) before the data reaches the AI model.
- **Strength**: Ensures high-fidelity input for the RAG model, significantly reducing hallucinations and improving the system's ability to handle complex, context-dependent queries.

6.3 2.3 Gap Analysis

Despite the many existing RAG frameworks and tools, there are considerable limitations to applying them in an enterprise context such as SCB TechX:

- **Limitation regarding Data Ingestion (The Ingestion Gap)**: Most open-source tools focus on the retrieval algorithm (the AI side) but neglect the ingestion complexity (the Data Ingestion side). There is a lack of robust pipelines designed to handle "messy" enterprise data formats while preserving semantic structure. Solving this requires specialized document parsing and OCR tools to extract high-fidelity content.
- **Limitation regarding Data Synchronization (The Synchronization Gap)**: Standard RAG implementations often treat knowledge bases as static datasets that require manual re-indexing. In active enterprise environments, data evolves constantly. There is a critical shortage of systems capable of Real-time Synchronization, where updates in the source (e.g., SharePoint) are instantly reflected in the AI's knowledge base without downtime. This requires a dedicated event-driven architecture using services like a Webhook Listener and a Message Queue (RabbitMQ).
- **Limitation regarding Context (The Contextual Gap)**: Standard vector search flattens data, losing the "relational" context (e.g., distinguishing between a draft and a final policy, or linking a meeting minute to its specific project). Current systems rarely combine vector search with relational filtering effectively. This gap requires implementing GraphRAG-style extraction to create a structured knowledge graph that preserves relationships between content chunks and documents.

6.4 2.4 Linkage to Project

This project addresses the identified limitations by focusing heavily on the Ingestion and Preparation Pipeline:

1. **Optimized Data Ingestion**: By designing a robust content extraction system, the project ensures that unstructured data from sources like Xhive/SharePoint is transformed into high-quality, queryable assets. This directly solves the "Ingestion Gap."
2. **Automated Knowledge Synchronization**: The project implements an automated pipeline that monitors source repositories for changes. This bridges the "Synchronization Gap" by ensuring that when files are added or modified in SharePoint, the AI's knowledge base is updated in near real-time without manual re-indexing.

3. **Hybrid Data Representation**: By researching and implementing a system that understands both the content and the structure/metadata of the files, the project addresses the "Contextual Gap," creating a semantic space that allows the AI to retrieve information with higher precision than standard flat searches.

To implement these solutions, the project takes a deep dive into the technical implementation of specific tools, including CrewAI for multi-agent reasoning, Qdrant for vector operations and relational filtering, and custom ETL pipelines utilizing advanced libraries for semantic and graph-based chunking, ensuring the foundational engineering is robust.

7 3. Proposed Solution / Design Concept

7.1 3.1 System Overview

The proposed system is an AI-powered Chatbot Application for searching and answering questions from a Document Database. Users can inquire with Natural Language through a Chat Interface. The system then processes the request in the background using Retrieval-Augmented Generation (RAG) technology combined with GraphRAG to retrieve relevant content from documents, both in terms of semantic similarity and structural data relationships. This retrieved content is used as context to generate accurate and complete answers. As a result, users receive answers quickly, referenced from the actual data in the documents, without having to spend time searching themselves.

In terms of architecture, the system consists of a Frontend and Backend responsible for supporting User Interaction in both the Chat Interface and Document Upload. It works in conjunction with 4 key services: SharePoint Webhook and File Storage Service, which listen for triggers from SharePoint when a file is uploaded, then pull and temporarily store the document before passing it to the Data Ingestion process. Data Ingestion extracts the text, performs Chunking, attaches Metadata, and sends it to the Embedding Service to convert the data into vectors for storage in the Vector Database. Finally, the AI Engine runs the main workflow, delegating tasks by using an LLM to generate a Search Query and perform data Retrieval from the database to synthesize an answer to be returned to the user.

7.2 3.2 Components

The system design adheres to the Modular Architecture principle, focusing on Decoupling functional components so that each Service is flexible and reusable with other

systems or applications in the future, not limited to use within this chatbot system. The various components are divided as follows

1. **Frontend Application**

   The user interface specifically for this chatbot system, developed with Next.js and React.

- **Responsibility:** Handles User Interaction through the Real-time Chat Interface, displays Citations, and manages the Document Upload screen.
- **Technology:** Next.js, React, Tailwind CSS

2. **Backend Orchestrator**

   Functions as an API Gateway and manages the Business Logic for this Application, developed with NestJS.

- **Responsibility:** Controls Session Management, stores Conversation History, and appropriately Routes requests from the Frontend to various destination Services.
- **Technology:** NestJS (TypeScript)

3. **AI Engine**

   A Multi-agent cognitive processing Module designed as a **Generic Reasoning Service**.

- **Responsibility:** A Multi-agent cognitive processing Module (CrewAI) designed as a Generic Reasoning Service. Chosen for its ability to define and configure complex multi-step workflows (agent roles) necessary for advanced retrieval techniques like GraphRAG and Selective Reading, providing Task Delegation or Reasoning services.
- **Technology:** Python FastAPI, CrewAI

4. **Embedding Service**

   A central Module for managing Vector Operations and Document Search.

- **Responsibility:** Provides services for converting text to Vector (Embedding) and searching data using Cosine Similarity. It is the sole Service with authority to manage the Qdrant Database, allowing other Applications to connect to deposit files or search data through a single API.
- **Technology:** Python FastAPI, Qdrant

5. Data Ingestion Service

A Module for Document Processing Pipeline.

- **Responsibility:** Acts as a General Purpose Pipeline for receiving files to perform ETL, Modality Detection (separating Text/Image/Table), Extraction (using robust document parsers), and advanced Chunking to prepare the data for use, regardless of the source. This is the component responsible for extracting relational metadata to support GraphRAG.
- **Technology:** Python FastAPI, RabbitMQ, OCR Tools

6. File Storage Service

A Module for a File Management System.

- **Responsibility:** Acts as an intermediary for storing and retrieving files from Object Storage (MinIO/S3). Supports Generating Presigned URL and notifying Events when a new file arrives, allowing other systems to use this Service for file management without connecting directly to S3 themselves.
- **Technology:** Python FastAPI, MinIO/S3

7. SharePoint Webhook Service

A Module for connecting External Integration Events.

- **Responsibility:** Functions as a Listener, waiting to receive file change Triggers from SharePoint and commands the subsequent Workflow. Can be immediately used with other projects that need to sync data from SharePoint.
- **Technology:** Python FastAPI, SharePoint Graph API

3.3 Constraints Considered

Based on the analysis of the system requirements and architecture, the development team has identified key technical constraints and challenges, and has defined design guidelines to address these issues as follows

1. Latency in Agentic Workflow

- **Problem:** The operation of the AI Engine in a Multi-agent System requires analytical reasoning and data transfer across multiple Agent steps, combined with data retrieval (RAG), resulting in a longer Response Time than typical Chatbots.

- **Planned Approach:** Design the system to work asynchronously for parts that do not require immediate results, and plan to use **Progress Streaming** to inform the user of the status of each Agent step in Real-time to reduce the perception of waiting.

2. **Context Granularity**

- **Problem:** Documents in the system are large and interconnected, but Language Models (LLMs) have limitations regarding the Context Window. Sending all information is not feasible, and coarse Chunking may lead to the AI misunderstanding the context.
- **Planned Approach:** In addition to normal Vector Search, we plan to use the **GraphRAG technique** to extract cross-file data relationships and will develop **Custom Agent Tools** (such as a *Sliding Window Reader* or *Page Retriever*). These tools allow the **CrewAI Agent** to dynamically and "selectively read" only the necessary, granular parts of the data instead of receiving a large block of information, optimizing the LLM's context utilization.

3. **Data Consistency**

- **Problem:** Source documents in SharePoint are constantly being edited, updated, or deleted. If the Vector Database system is not well-managed, it will lead to issues with junk data or outdated knowledge.
- **Planned Approach:** The Data Ingestion system is designed to support **CRUD operations** linked to the **SharePoint Webhook**. The **RabbitMQ message queue** is used to guarantee reliable and ordered processing of these update events to ensure that when the source file changes, the system automatically updates or deletes the corresponding data in the Vector Store.

4. **LLM Safety & Guardrails**

- **Problem:** Allowing users to converse with the LLM carries security risks, such as Prompt Injection attacks or the model providing inappropriate answers.
- **Planned Approach:** An **AI Guardrails** layer will be installed to check and filter messages on both the input (Input Validation) and output (Output Validation) sides to prevent harmful commands and control the content of the response within the defined scope.

5. **Complex Document Structure and Thai Language**

- **Problem:** Extracting information from documents containing tables, images, and Thai word segmentation is complex and prone to high error rates.
- **Planned Approach:** Plan to use the **Modality Detection process** (in the Data Ingestion Service) to distinguish content types before entering the extraction process and select a **Multilingual-compatible Embedding Model** to enhance the search performance for Thai language.

6. External Dependencies

- **Problem:** The system relies on SharePoint and LLM services (OpenAI/Anthropic). If these services fail, the main system will be affected.
- **Planned Approach:** Utilize the **Modular Architecture** design to separate functional components and use internal File Storage as a data buffer to reduce constant direct connection to SharePoint, which helps mitigate the risk if the source system experiences temporary issues.

7.3 3.4 Diagrams

Architecture diagrams

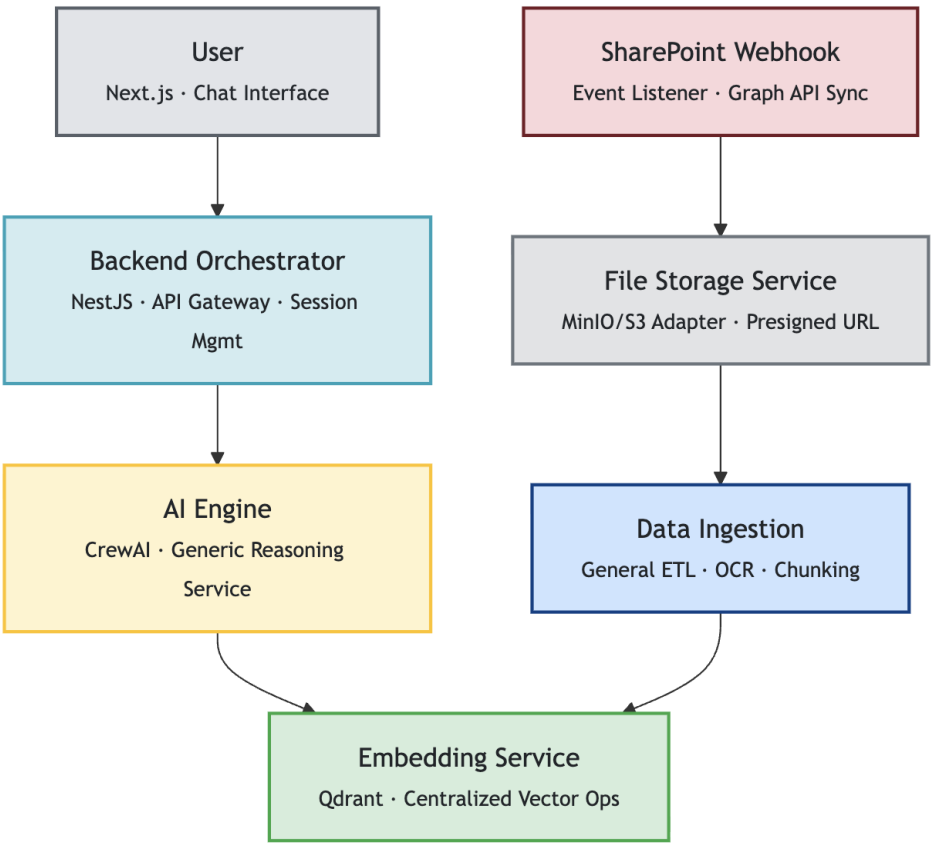
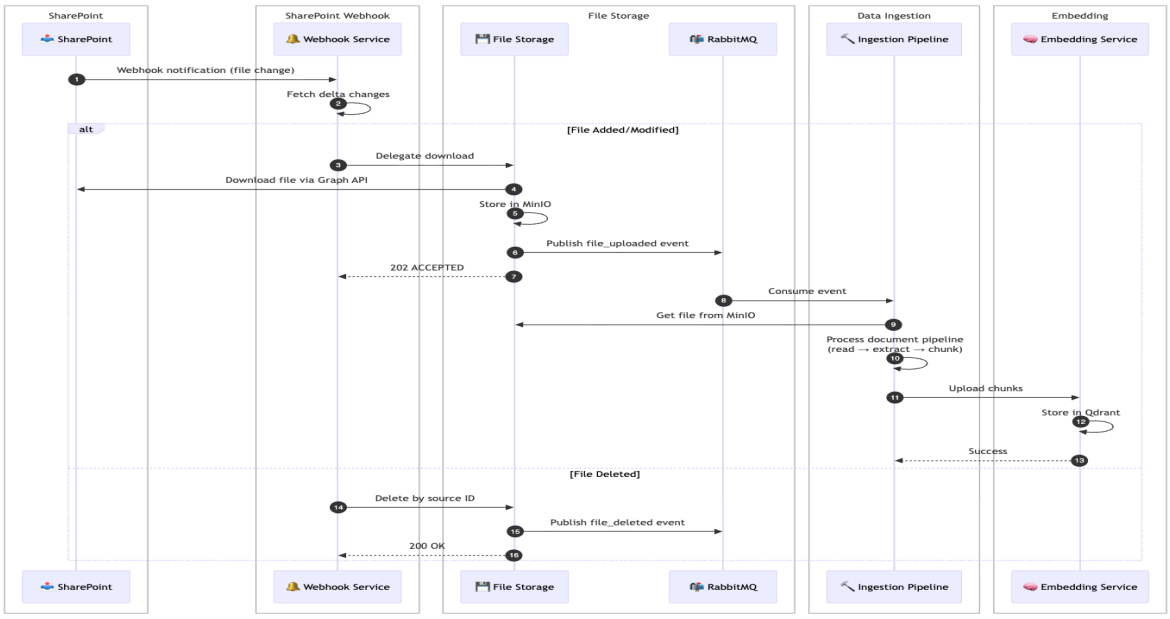


Fig 1. C4 Level 2 Diagram of the AiQ System

Workflows



Fic 2. Sequence Diagram of RAG & Agentic

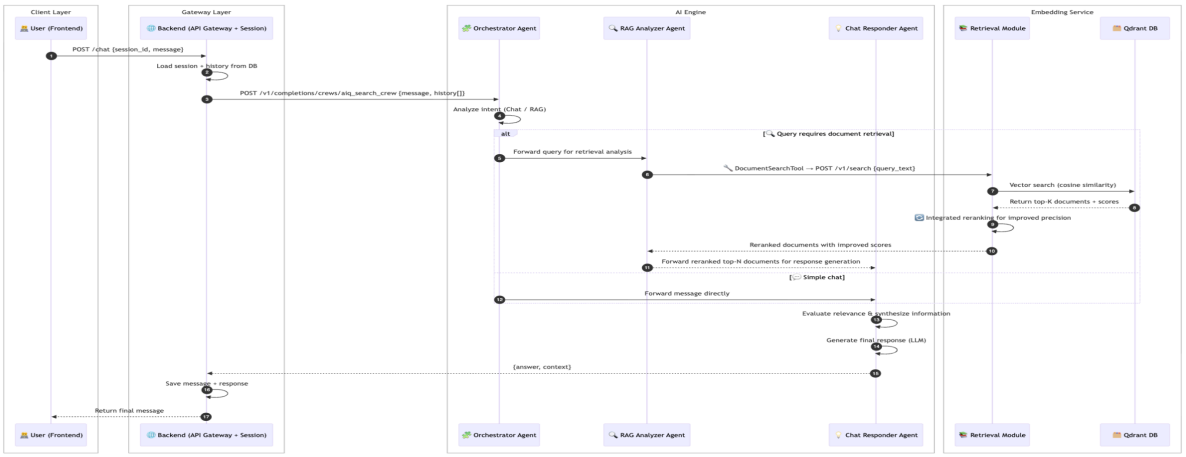


Fig 3. Sequence Diagram of the Data Ingestion Pipeline

8 4. Teamwork Plan

8.1 4.1 Roles and Responsibilities

1. **Data Engineer:** Designs and develops the data pipeline for extracting and transforming information from multiple document formats (PDF, DOCX, PPTX, images). Ensures data quality, proper storage in the database, and readiness for use in the RAG system.
2. **Project Manager:** Manage project structure and timeline, communication across team members, track progress, and ensure that all results meet objectives.
3. **Web Developer:** Develop a web application, including a document upload system, Chat UI, and integrations with backend APIs. Also designs UX/UI for the web.
4. **AI Engineer:** Responsible for designing and developing AI components such as AI Agents, the RAG system, retrieval pipelines, and pipeline optimization to ensure accurate, efficient, and reliable performance for the AiQ platform.

8.2 4.2 Collaborative Plan

Scrum approach with bi-weekly sprints, where tasks are clearly defined and tracked through Jira. The team conducts stand-up meetings 2–3 times per week to maintain visibility of progress and to identify and resolve blockers early.

8.3 4.3 Task Coverage

1. **Data ingestion & Processing:** Data Engineer
2. **AI Components:** AI Engineer
3. **Backend API & Frontend Interface:** Fullstack Developer
4. **Coordination, Documentation, Stakeholder Communication:**Project Manager

8.4 4.4 Workload Balance

Work is distributed according to each member's specialty. Story points are assigned to ensure tasks are fairly distributed across each sprint (following Scrum principles), so that every team member receives an appropriate amount of work.

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10 5. Timeline

11 Semester 1

Week	Task / Milestone	Deliverable
Week 1–2	Conduct literature review and study relevant background materials	Review related work and draft the background, problem statement, and project scope
Week 3–4	Gather requirements from users and stakeholders	Define project objectives, and write epics, user stories, and acceptance criteria
Week 5–7	System-level architecture design, tool selection, model selection, and data flow design	Draft architecture diagram, technology/tool stack list, model selection justification
Week 8–10	Develop end-to-end web application (UI, backend APIs, database) without data ingestion service	Functional prototype of the web application & API documentation
Week 11–12	Investigate, design, and prototype Data Ingestion Service (pipeline, ETL/ELT approach, orchestration)	Data ingestion service prototype & ingestion workflow diagram
Week 13–14	Integrate and build end-to-end Data Ingestion Service with the existing web application	Fully integrated data ingestion pipeline & system integration test report
Week 15-16	Final proposal presentation, documentation, and refinement	Final report & final slides

12 Semester 2

Week	Task / Milestone	Deliverable
Week 1–2	Enhance ingestion pipeline, optimize chatbot performance, and run initial evaluations	Updated ingestion pipeline, baseline performance metrics, initial evaluation report
Week 3–4	Further optimization: ingestion improvements, better accuracy & lower latency, continued evaluations	Optimized ingestion v2, improved model performance metrics, mid-phase progress report
Week 5–7	Build and refine key web features (citations, SSO, UI/UX improvements)	Completed core web functionality, demo-ready feature build
Week 8–10	Product polishing: bug fixing, refinement, readiness for User Acceptance Testing (UAT)	Pre-UAT build, UAT checklist & test cases
Week 11–12	Conduct UAT and begin final report	UAT results, defect list, draft final report
Week 13–14	Resolve UAT issues, performance stabilization	Resolved issue log, final system build
Week 15-16	Deliver final report and presentation	Final written report, presentation slide deck, final demo

13 6. Ethics, Privacy, and Professional Considerations

13.0.1 6.1 Credibility of Information

Our project establishes data credibility by grounding all work in dependable and well-validated materials. Internal documents from SCB TechX provide accurate domain knowledge that reflects real workflows and requirements. External information such as academic papers and official documentation for tools and frameworks are carefully selected from reliable sources. Together, these materials form a solid and reliable knowledge base for the project.

6.2 Intellectual Property & Law

All activities under this project adhere to confidentiality obligations as defined in the Capstone Non-Disclosure Agreement (NDA) between SCB Tech X and participating students. All proprietary information including technical data, business information, processes, software, and internal documentation remains the exclusive property of SCB Tech X and must not be disclosed, reproduced, or used beyond the scope of the Capstone project. Students must store confidential data securely, restrict access only to authorized project members.

14 References

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16 Appendices

16.1 Wireframes and Mockups

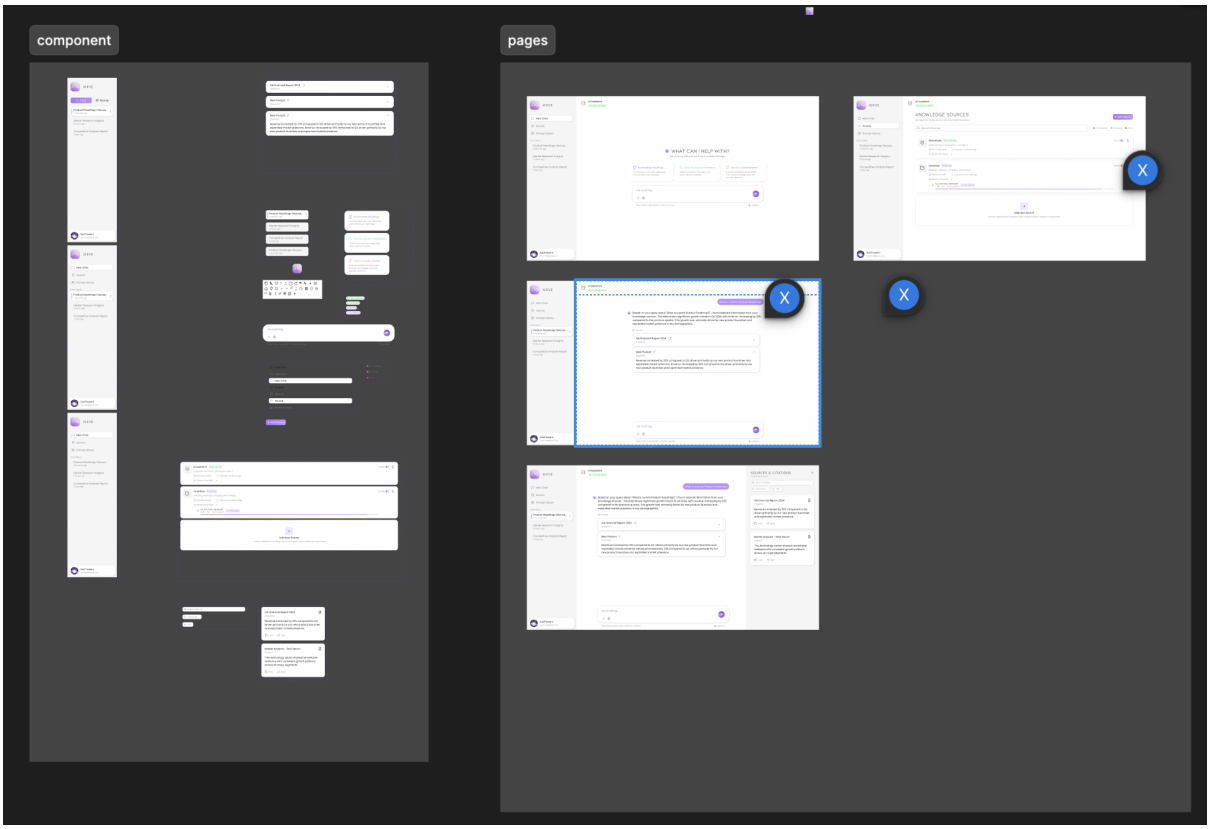

Figure 1: Figma components as our design system

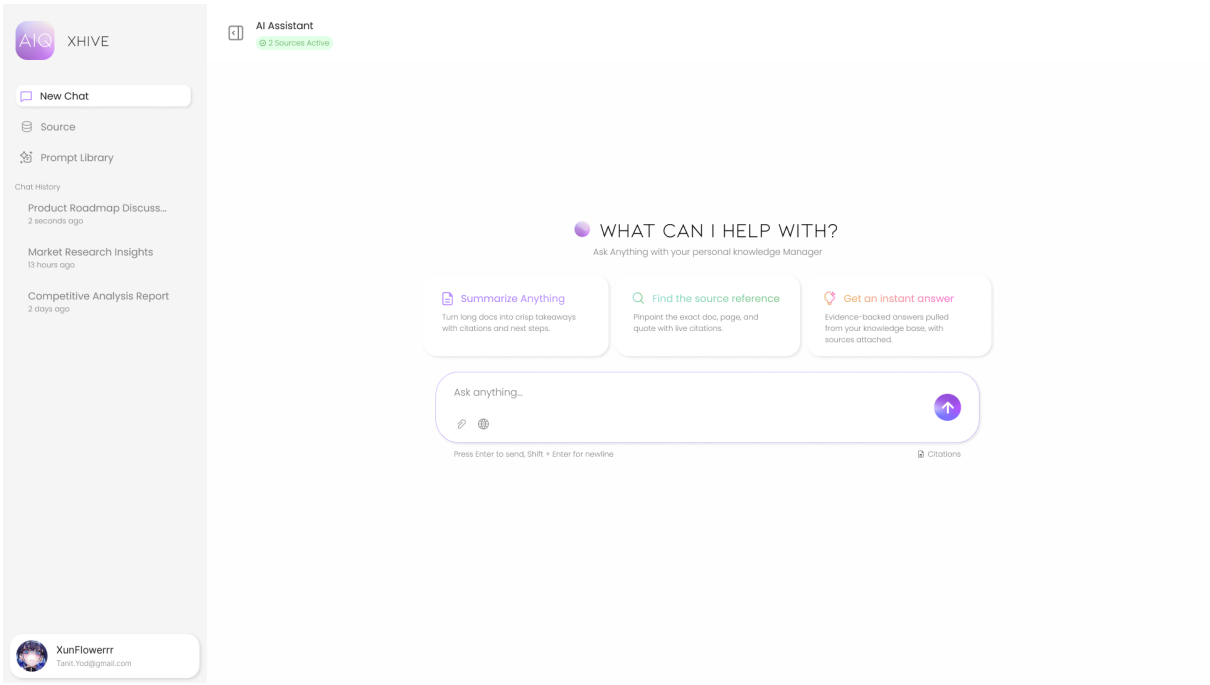

Figure 2: Web chat interface for demonstrating system intelligence

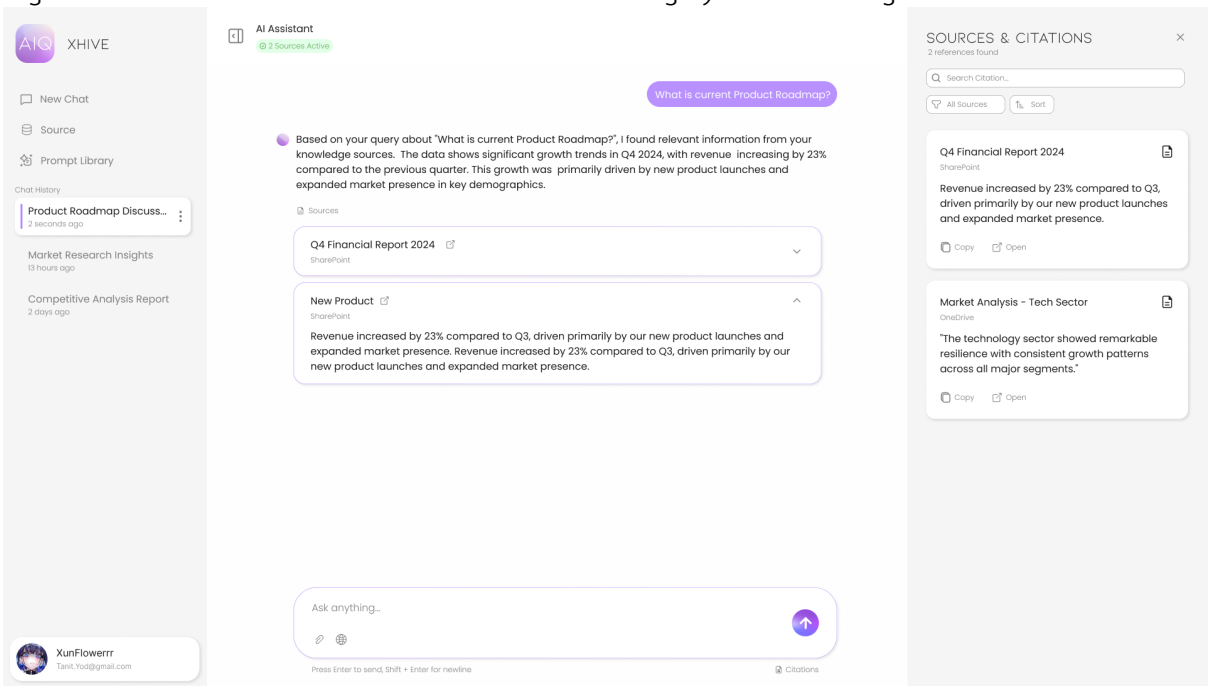

Figure 3: Web chat interface for viewing citation details

16.2 Early Test Datasets

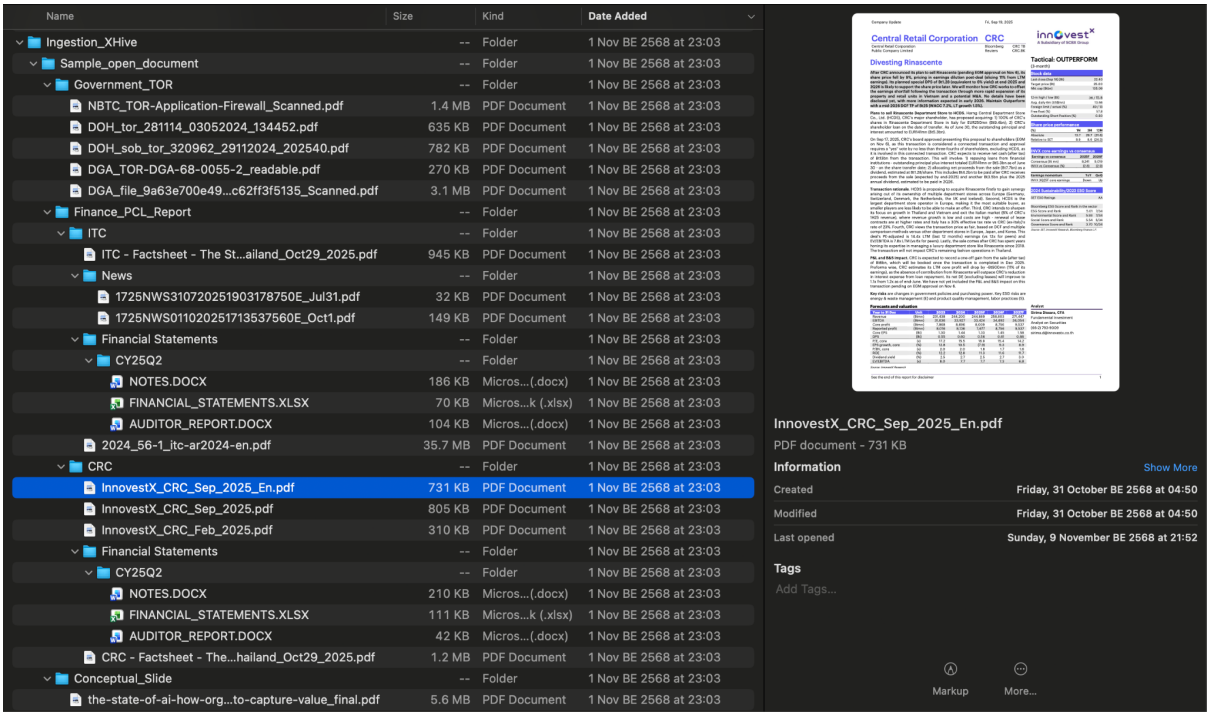

Figure 1: Example of test dataset